

PRODUCT REQUIREMENTS DOCUMENT

Automate Workflows

A visual workflow-automation platform (Zapier / Pabbly-class) for the Automate product suite

Version 1.0 • Draft for Review

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1. Document Overview

This PRD defines the product for “Automate Workflows” — a no-code / low-code workflow-automation builder, in the same category as Zapier, Pabbly Connect, and Make. It lets users connect apps with triggers and actions to automate repetitive, cross-app business processes, and sits inside the same product family as Automate Chats (WhatsApp marketing) and Automate Forms (form builder), so those products can become native trigger/action nodes.

The PRD is organized so engineering, design, and growth can each read the sections relevant to them. Feature scope is phased from MVP through to a full-featured, AI-enabled, enterprise-ready product, with information architecture, core user flows, and non-functional requirements included for every phase.

1.1 How to read this document

- Section 4 (Personas) and 5 (IA) ground every later decision — read these first.
- Section 7 (Feature Roadmap) is the master scope list; Section 8 expands MVP features into detailed specs with acceptance criteria.
- Section 6 (Core Concepts) defines the vocabulary used everywhere else — keep it consistent in UI copy and engineering naming.

2. Problem Statement & Vision

2.1 Problem

Businesses use 10–50+ SaaS tools (CRM, forms, chat, payments, sheets, email) that don't talk to each other out of the box. Teams either hire developers to write glue-code integrations, or waste hours on manual copy-paste between tools. Existing automation platforms solve this but have gaps our users feel directly:

- Zapier is powerful but expensive at scale — every internal step (filter, formatter, router) consumes a billable task.
- Pabbly is cheap and has unlimited workflows, but its editor, error-handling UX, and documentation feel dated and thin for complex logic.
- Neither is natively integrated with WhatsApp-first marketing workflows or form-triggered automations the way our own products (Automate Chats, Automate Forms) need to be.

2.2 Vision

Give any business user — not just developers — a visual canvas to connect Automate Chats, Automate Forms, and thousands of third-party apps into reliable, observable, multi-step workflows, priced fairly (internal logic steps are free, like Pabbly) with an editor experience as polished as Zapier's, and a first-class AI assistant that can build and explain workflows in plain language.

2.3 Goals (Year 1)

- Ship an MVP that reliably automates the top 20 use cases our existing Automate Chats / Automate Forms customers already ask for.
- Reach parity with Pabbly Connect's core feature set (triggers, actions, multi-step, filters, routers, delay, webhooks) by end of Phase 2.
- Reach feature parity with Zapier's mid-tier plan (paths, formatter, error handling, team collaboration) by end of Phase 3.
- Differentiate with native Automate-suite nodes, transparent "internal steps are free" pricing, and an AI workflow builder by Phase 4.

2.4 Non-goals (for now)

- Building our own iPaaS-grade enterprise data warehouse / ETL product.
- Full autonomous AI agents (Zapier Central-style) — tracked as a post-Phase-4 exploration, not committed scope.
- On-premise / self-hosted deployment.

3. Market & Competitive Snapshot

Both reference products validate the category and show two different winning strategies: Zapier wins on breadth (7,000+ app integrations) and polish, layering AI (Copilot, AI Actions, Canvas) on top of a mature core; task-based pricing is its biggest friction point. Pabbly Connect wins on price and simplicity — unlimited workflows, only external actions billed as tasks, lifetime-deal pricing — but is seen as less capable for advanced branching logic and has thinner docs.

Dimension	Zapier	Pabbly Connect	Automate Workflows (target)
Pricing model	Every step (incl. filters/formatters) consumes a task	Only external actions billed; internal logic steps free	Only external actions billed; unlimited workflows on paid tiers
App coverage	7,000+ apps	1,000–2,000+ apps	MVP: ~30–50 popular apps + native Automate suite; grow via community + generic HTTP/webhook connector
Builder UX	Polished drag-and-drop, Canvas AI planner	Functional but dated, thinner docs	Polished visual canvas from day one; AI assist from Phase 4
Advanced logic	Paths, sub-Zaps, formatter, loops	Routers, iterators, JS/Python code steps	Router/branch, filter, delay/schedule, loop, code step (Phase 2–3)
AI capabilities	Copilot, AI Actions, Central agents, MCP	Basic AI assistant	AI workflow generation + AI action step (Phase 4)

Implication for scope: MVP should not try to out-integrate Zapier’s app catalog. It should nail the builder experience, reliability, and pricing fairness (Pabbly’s model), for a focused app catalog anchored on the Automate product suite plus the most-requested third-party tools, then expand breadth and add AI in later phases.

4. Target Users & Personas

Persona A — “Riya, Growth/Marketing Ops Lead”

- Runs WhatsApp campaigns in Automate Chats and lead forms in Automate Forms; not a developer.
- Wants: new form submission → send WhatsApp welcome message → add to CRM → notify sales on Slack.
- Cares about: reliability, clear error alerts, not blowing through a task quota.

Persona B — “Arjun, Agency Owner / Solopreneur”

- Manages automations for 5–10 clients; price-sensitive, wants unlimited workflows and generous task limits.
- Wants: templates he can clone per client, and simple sharing/permissions so clients can view (not edit) their own workflow.

Persona C — “Neha, Technical Ops / RevOps”

- Comfortable with JSON, webhooks, basic scripting.
- Wants: branching logic, custom code steps, retry/error handling, and an audit log for debugging failed runs.

Persona D — “IT/Admin, Mid-market Buyer” (Phase 3–4)

- Needs SSO, role-based access, usage analytics per team, and data-residency / security assurances before rollout.

5. Information Architecture

5.1 Top-level navigation (app shell)

- Home / Dashboard — workflow health overview, recent runs, quick-create.
- Workflows — list of all workflows (grid/list, search, filter by status/app/folder).
- → Workflow Editor (canvas) — opened per workflow.
- → Workflow Run History / Logs — opened per workflow.
- Connections — manage authenticated app accounts (add/remove/reauthorize).
- Templates / Gallery — pre-built workflow templates by category and by app.
- Apps Directory — browse available triggers/actions per app.
- Team — members, roles, shared folders (Phase 2+).
- Usage & Billing — plan, task consumption meter, invoices, upgrade.
- Settings — account, notifications, API keys, security (SSO in Phase 3+).
- Help / Docs — in-app guides, contextual tooltips, support chat.

5.2 Workflow Editor structure (the core screen)

- Canvas (center) — vertical step list (Zapier-style) with branch/path support from Phase 2; each step is a node: Trigger, Action, Filter, Router, Delay, Code, AI Action.
- Step configuration panel (right drawer) — opens when a step is selected: choose app → choose event → connect account → map fields → test step.
- Top bar — workflow name, on/off toggle, Test/Run button, version history, share/publish.
- Left rail (optional, Phase 2+) — mini-map for large multi-branch workflows.
- Bottom drawer — test run output / execution data inspector per step.

5.3 Site map (simplified)

Dashboard → Workflows → [Workflow Editor, Run History] | Connections | Templates | Apps Directory | Team | Usage & Billing | Settings | Help

5.4 MVP Screen Inventory

22 screens across 7 categories cover the full MVP scope defined in Section 7 (Phase 1). “Screen” includes full pages and major modal/drawer states that a designer needs to draw separately — it does not count minor variations (loading/empty/error states) as separate screens unless the layout materially changes.

A. Authentication & Onboarding (5 screens)

#	Screen	Detailed description
1	Sign Up	Email + password fields plus a “Continue with Google” button. Inline validation on email format and password strength. Links to Log In and to Terms/Privacy. On submit, creates the account and routes to Verify Email.

#	Screen	Detailed description
2	Log In	Email + password fields, “Continue with Google” button, “Forgot password?” link. Shows an inline error banner for invalid credentials without specifying which field is wrong (security best practice).
3	Verify Email	Post-signup interstitial: “We sent a link to {email}” with a Resend button (rate-limited) and a Change email link. User cannot publish a workflow until this is completed; the rest of the app remains usable in draft mode.
4	Forgot / Reset Password	Two-step: (a) email entry to trigger the reset email, (b) new-password form reached via the emailed link, with confirm-password field and strength meter.
5	Onboarding — “What do you want to automate?”	One-time screen shown right after verification: 4–6 large tappable cards mapped to starter templates (e.g. “Form → WhatsApp”, “WhatsApp → CRM”, “Start from scratch”). Selecting a card opens the Workflow Editor pre-loaded with that template.

B. Dashboard (1 screen)

#	Screen	Detailed description
6	Dashboard (Home)	Landing screen after login: workflow health summary (active/paused/error counts), a feed of recent run activity across all workflows, and a prominent “Create Workflow” button. Empty state (no workflows yet) replaces the feed with the onboarding template cards from Screen 5.

C. Workflows & the Editor (5 screens)

#	Screen	Detailed description
7	Workflows List	Grid or list of every workflow the user owns or has access to, with name, status toggle (On/Off/Draft), last-run status icon, and last-modified date. Includes search and filter by status/app. Row click opens the Workflow Editor; a status-icon click deep-links to Run History.
8	Workflow Editor (Canvas)	The core screen (detailed in Section 5.2): vertical step list, top bar with name/on-off toggle/Test button, “+” affordance to add steps. This is the screen the user spends the most time in and should be designed first.
9	Step Configuration Panel	Right-hand drawer that opens over the canvas when a step is added or selected: app picker → event/action picker → connection picker (or “Connect new”) → field-mapping form with an inline variable picker. Closing the drawer returns focus to the canvas without losing unsaved input.
10	Test Step / Data Inspector	Bottom drawer shown after “Test step” is clicked: displays the raw JSON request/response for that step in a collapsible tree view, with a plain-language summary line (“Success — 3 fields returned” or the specific error).

#	Screen	Detailed description
11	Publish Validation Summary	Inline panel/modal shown if the user tries to turn a workflow On while required fields are unmapped: a scrollable list of the specific steps/fields at fault, each linking directly back to that step in the editor.

D. Run Monitoring (2 screens)

#	Screen	Detailed description
12	Run History	Per-workflow list of executions: timestamp, trigger source, overall status (success/error/queued), and duration. Filterable by status and date range. Empty state reads “Waiting for the next trigger event...” for a workflow that’s live but hasn’t fired yet.
13	Run Detail	Opened from a Run History row: a vertical breakdown of every step in that run with its individual status, plus the same input/output JSON viewer as Screen 10. Includes the “Retry from this step” action on the failed step.

E. Connections (2 screens)

#	Screen	Detailed description
14	Connections Manager	List of every authenticated app account: app icon/name, connected account label (e.g. an email or workspace name), status badge (Active / Needs reauthorization), and last-used date. Actions per row: reauthorize, rename, disconnect.
15	Add / Edit Connection	Modal that adapts to the app’s auth type (Section 8.8): for OAuth apps it shows a single “Continue to {App}” button that hands off to the provider’s consent screen; for API-key apps it shows the specific labeled fields that app needs (e.g. “API Key”, “Domain”) with inline validation against a live test call before saving.

F. Templates & Apps Directory (3 screens)

#	Screen	Detailed description
16	Templates Gallery	Browsable, searchable grid of the 15–20 curated MVP templates, filterable by category (Lead capture, CRM sync, Notifications, etc.) and by app used. Each card shows the trigger → action chain at a glance.
17	Template Preview	Opened from a gallery card: shows the full step sequence read-only, a short description of the use case, and a “Use this template” button that clones it into a new draft workflow in the Editor.
18	Apps Directory	Browsable list of all MVP-supported apps (Section 8.8) with their available triggers and actions, searchable by app name. Used both for pre-purchase evaluation and as an entry point into starting a new workflow from a specific app.

G. Billing, Account & Settings (4 screens)

#	Screen	Detailed description
19	Usage & Billing	Current plan name and limits, a progress-bar-style meter of tasks consumed this month vs. the plan cap, and a table of past invoices. Shows the 80%/100% warning banners described in Section 9.4.
20	Plan Upgrade / Checkout	Plan comparison cards (Free/Starter/Pro/Business per Section 8.7) with a payment form for card details. Confirmation state shows the new limits taking effect immediately and, if reached via the usage banner, resumes any queued runs.
21	Account Settings	Profile fields (name, email, password change), connected Google account status, and account deletion. Standard settings-page layout with saved-state confirmation toasts.
22	Notification Settings	Toggles for which events send an email (run failed, task-limit warning, teammate invited) and, where applicable, a Slack-alert-channel picker for the same event types.

Not counted as separate screens: loading states, hover/focus states, and confirmation toasts — these are states of the screens above, not additional screens to design. Team/role-management screens are Phase 2 scope (Section 7) and excluded from this MVP count.

6. Core Concepts & Terminology

Term	Definition
Workflow	A saved automation made of one Trigger and one or more subsequent Steps.
Trigger	The event that starts a workflow run (e.g. “New form submission”, “New WhatsApp message”, “Schedule”, “Webhook received”).
Action	A step that performs an operation in an app (e.g. “Send WhatsApp template”, “Create CRM contact”).
Step	Any node on the canvas — Trigger, Action, Filter, Router, Delay, Code, or AI Action.
Connection	An authenticated account link to a third-party app (OAuth, API key, etc.), reusable across workflows.
Filter	A condition that stops the run unless met (internal step, not billed).
Router / Path	Branches a workflow into multiple conditional paths from one trigger.
Delay / Schedule	Pauses execution for a duration or until a specific time.
Task	One billable unit — one successful execution of one external Action. Internal steps (filter, router, delay, formatter) are free.
Run / Execution	One end-to-end pass of a workflow for a single trigger event, with a per-step log.
Template	A pre-built, cloneable workflow for a common use case.

7. Feature Roadmap — MVP to Final Product

Four phases. Each phase is shippable and usable on its own; later phases build on earlier ones without rework.

Phase 1 — MVP (Launch)

Goal: prove the core loop — connect two apps, build a reliable multi-step workflow, see it run — for our existing Automate Chats / Automate Forms user base.

- Account & auth (signup, login, email verification, password reset).
- Connections manager: OAuth + API-key connections for ~15–20 launch apps incl. Automate Chats, Automate Forms, Google Sheets, Gmail, Slack, WhatsApp Cloud API, Webhooks, Shopify, HubSpot, generic HTTP request.
- Workflow editor: single linear path, Trigger + up to 10 Actions, field mapping with variable picker, per-step “test step” with live sample data.
- Trigger types: app event (polling/webhook-based), Schedule, Manual/Webhook-catch.
- Filter step (conditional stop) — free, not billed.
- Turn workflow on/off; draft vs published state.
- Run history: per-run status (success/error), per-step input/output payload viewer.
- Basic error handling: failed step marks run as failed, sends email notification to owner; manual retry button.
- Templates gallery: 15–20 curated templates seeded around Automate Chats + Automate Forms use cases.
- Usage meter: tasks consumed this month vs plan limit, with in-app warning at 80%/100%.
- Pricing/plans: Free (limited tasks, limited workflows), Starter, Pro — unlimited workflows on all paid tiers; internal steps free.
- Responsive web app (desktop-first; mobile view is read-only dashboard/status).

Phase 2 — Growth

Goal: match Pabbly-class core automation depth and add team collaboration.

- Router/Path step: branch a workflow into multiple conditional paths.
- Delay step: fixed delay or delay-until-datetime.
- Formatter step: text/number/date transforms, no-code data mapping helpers.
- Iterator/Loop step: run subsequent actions once per item in a list.
- Expanded app catalog (100+ apps), community-requested app request form.
- Team workspaces: invite members, roles (Owner/Editor/Viewer), shared folders for workflows.
- Version history with rollback for a workflow.
- Improved error handling: configurable retry policy per step, “continue on error” option, Slack/email alert routing.
- Search across workflows, connections, and run logs.
- Public workflow template marketplace (submit + browse community templates).

Phase 3 — Scale

Goal: reach Zapier mid-tier parity and become enterprise-credible.

- Code step: custom JavaScript/Python snippet for advanced data transforms.
- Sub-workflows: call one workflow from another (reusable automation blocks).
- Advanced field mapping: nested/array data, conditional field values.
- SSO (SAML/OAuth) and role-based access control at org level.
- Audit log for compliance (who changed/ran what, when).
- Rate-limit-aware execution (auto-throttling per destination app).
- Multi-environment support: staging vs production workflow copies.
- SLA-grade reliability: 99.9% uptime target, status page, incident notifications.
- Advanced analytics dashboard: task usage by workflow/app/team, cost forecasting.
- Public API + developer platform for partners to build custom app integrations.

Phase 4 — AI & Enterprise (Final Product)

Goal: differentiate with AI-assisted building and enterprise depth.

- AI workflow generator: describe an automation in plain language, get a draft workflow to review and connect.
- AI Action step: summarize, classify, extract, translate, or generate content inline in any workflow, model-agnostic.
- AI-assisted field mapping and error diagnosis (“why did this step fail, and how do I fix it”).
- Visual workflow planner (map multi-workflow architecture before building, like a lightweight Canvas).
- Data store (lightweight tables) to hold state between workflow runs without a third-party database.
- Enterprise controls: data residency options, custom SLAs, dedicated support, SCIM provisioning.
- White-label / embedded automation: let Automate Chats and Automate Forms customers build simple automations inside those products, powered by this engine.

8. Detailed Feature Specifications — MVP

8.1 Account & Authentication

Description

Email/password and Google OAuth signup; email verification required before first workflow can be published; standard forgot/reset password flow.

Acceptance criteria

- User can sign up with email+password or Google in under 60 seconds.
- Unverified accounts can build workflows in draft but cannot turn them on.
- Password reset email arrives within 60 seconds and link expires in 24 hours.

8.2 Connections Manager

Description

A central place to authenticate and store credentials for each third-party app, reusable across any workflow.

Acceptance criteria

- User can add a connection via OAuth (redirect + callback) or API key/secret form, per app requirements.
- Connections list shows app, account name/email, status (Active/Needs reauthorization), and last-used date.
- Revoking or reauthorizing a connection does not delete workflows that use it; affected workflows show a “reconnect needed” badge.
- Credentials are encrypted at rest; never displayed in plaintext after initial entry.

8.3 Workflow Editor

Description

Visual, vertical step-list canvas. User picks a Trigger, then adds Action/Filter steps below it in sequence.

Key interactions

1. Click “Create Workflow” → choose a Trigger app and event (or start from a template).
2. Connect or select an existing connection for that app.
3. Configure trigger fields (e.g., which form, which WhatsApp number) → fetch and show sample trigger data.
4. Click “+” to add an Action step → choose app + event → map fields using the sample data (drag-and-drop variable picker or type ‘/’ to insert a variable).
5. Test the step → see live output → repeat for each subsequent step.
6. Name the workflow, toggle it On — workflow moves from Draft to Published/Live.

Acceptance criteria

- Field mapping supports inserting any upstream step’s output field as a variable into any downstream field.

- “Test step” never sends a real action-side effect during trigger/mapping tests unless the user explicitly runs “Test & continue” on an Action step.
- A workflow cannot be turned on if any required field on any step is unmapped — inline validation blocks publish with a clear error list.
- Undo/redo within an editing session; autosave draft every change.

8.4 Trigger Types

Type	Behavior
App event (polling)	Checks the source app on an interval (e.g., every 1–5 min depending on plan) for new matching data.
App event (webhook/instant)	Source app pushes data to us in real time where the app supports webhooks (incl. Automate Chats, Automate Forms).
Schedule	Runs on a cron-like schedule (every X minutes/hours, specific days/times).
Webhook catch	Generates a unique URL; any external system can POST data to start the workflow.

8.5 Filter Step

A condition (field, operator, value; AND/OR grouping) that stops the run silently if not met. Free — does not consume a task.

8.6 Run History & Debugging

Acceptance criteria

- Every run shows: timestamp, trigger source, overall status, duration, and a per-step breakdown.
- Clicking a step shows raw input payload and raw output payload (JSON viewer, collapsible).
- Failed runs show the error message returned by the destination app in plain language plus the raw error.
- User can manually re-run a failed execution from the exact point of failure without repeating already-successful steps.

8.7 Usage Meter & Billing

Description

Only successful external Actions count as tasks (Filters, Routers, Delay, Formatter are free). Meter resets monthly on the billing date.

Plans (draft — to be finalized with pricing/finance)

Plan	Price	Tasks/mo	Workflows	Steps per workflow
Free	\$0	500	3	Up to 5

Plan	Price	Tasks/mo	Workflows	Steps per workflow
Starter	\$15/mo	5,000	Unlimited	Up to 15
Pro	\$39/mo	20,000	Unlimited	Unlimited
Business	Custom	Custom / pooled	Unlimited	Unlimited + team seats

8.8 MVP App Connections — Full Configuration Specification

18 launch connections, chosen to cover the workflows our existing Automate Chats / Automate Forms customers already run, plus the generic connectors that let users self-serve anything not yet natively supported. Each entry below defines exactly how the connection authenticates, what it can trigger on, what actions it exposes, and how data is synced (webhook = instant, polling = interval-based check).

Group 1 — Native Suite

1. Automate Chats

Auth type	Internal SSO — same login session as the user's Automate Chats account; no external OAuth or API key needed.
Setup steps	Click Connect → auto-authorized → select the specific WhatsApp Business number/account to use.
Triggers	New WhatsApp message received; Message status changed (sent/delivered/read/failed).
Actions	Send WhatsApp template message; Send WhatsApp session message; Add/update contact tag.
Sync mode	Webhook — native, instant (shares the same event bus as Automate Chats).
Notes	No token refresh required; account list is fetched live from the shared user session.

2. Automate Forms

Auth type	Internal SSO, same as above.
Setup steps	Click Connect → select the specific form from a live dropdown of the user's forms.
Triggers	New form submission (with full field payload).
Actions	None at MVP (source-only connector).
Sync mode	Webhook — native, instant.
Notes	Field schema for mapping is pulled directly from the form builder's field definitions.

Group 2 — Lead Capture & Forms

3. Google Forms

Auth type	OAuth 2.0 via Google (scopes: forms.responses.readonly, drive.readonly).
Setup steps	Google consent screen → grant access → pick account → select a specific form from a dropdown populated via the Drive/Forms API.
Triggers	New form response submitted.
Actions	None at MVP (source-only connector).
Sync mode	Polling, every 1–5 minutes depending on plan (Google Forms has no native push webhook).
Notes	Requires a verified Google Cloud OAuth consent screen and API quota project before launch.

4. Typeform

Auth type	OAuth 2.0 (Typeform Connect) or Personal Access Token as fallback.
Setup steps	OAuth redirect → authorize → select a specific Typeform from the account's form list.
Triggers	New response submitted.
Actions	None at MVP (source-only connector).
Sync mode	Webhook — we register a Typeform webhook automatically via their API on connect; instant.
Notes	Webhook auto-deregistered if the user disconnects or deletes the workflow.

Group 3 — CRM / Sales

5. HubSpot

Auth type	OAuth 2.0 (HubSpot App Marketplace flow); scopes: crm.objects.contacts (read/write), crm.objects.deals (read/write).
Setup steps	OAuth consent → select HubSpot portal (if user has multiple).
Triggers	New contact created; Deal stage changed.
Actions	Create/update contact; Create deal; Add note or timeline event to a contact.
Sync mode	Webhook where HubSpot's subscription supports it; polling fallback every 5 minutes for lower tiers.
Notes	Field mapping pulls the portal's actual custom-property list, not just standard fields.

6. Google Sheets

Auth type	OAuth 2.0 via Google (scopes: spreadsheets, drive.file).
Setup steps	Google consent → pick account → select spreadsheet and specific sheet/tab via a file picker.
Triggers	New row added.
Actions	Add row; Update row (lookup by column value); Lookup row (for enrichment steps).
Sync mode	Polling for the trigger (1–5 min, no native push); actions write in real time.
Notes	Column headers from row 1 are auto-read to build the field-mapping list.

7. Pipedrive

Auth type	API token + company domain at MVP (OAuth 2.0 added in Phase 2).
Setup steps	User pastes their Pipedrive API token and company subdomain.
Triggers	New deal created; Deal stage changed.
Actions	Create/update person; Create deal; Add activity or note.
Sync mode	Webhook — auto-registered via Pipedrive’s webhooks API on connect; instant.
Notes	Token is validated with a test call at connect time so bad tokens fail fast, not on first run.

Group 4 — Communication / Notifications

8. Slack

Auth type	OAuth 2.0 via Slack app install; scopes: chat:write, channels:read, users:read.
Setup steps	“Add to Slack” OAuth button → pick workspace → choose channel from a live dropdown.
Triggers	None at MVP (action-only connector; message-trigger moves to Phase 2).
Actions	Send channel message; Send direct message; Post formatted/block message.
Sync mode	Instant — actions call Slack’s Web API directly.
Notes	Channel and user lists refresh on demand so newly created channels appear without reconnecting.

9. Gmail

Auth type	OAuth 2.0 via Google (scopes: gmail.send, gmail.readonly).
Setup steps	Google consent → grant access to the specific mailbox.
Triggers	New email received matching a label/filter.

Actions	Send email (with optional attachment).
Sync mode	Polling for the trigger at MVP (1–5 min); push-based Gmail watch is a Phase 2 upgrade.
Notes	Sending uses the authenticated user's own address as sender to avoid spam-flagging.

10. Telegram

Auth type	Bot API token (user creates a bot via @BotFather and pastes the token).
Setup steps	Paste bot token → user adds the bot to their channel/group or starts a chat with it.
Triggers	New message sent to the bot.
Actions	Send message via the bot.
Sync mode	Webhook — we call Telegram's setWebhook on connect; instant.
Notes	Lightweight alternative to Slack for smaller teams; no OAuth screen needed.

Group 5 — Commerce

11. Shopify

Auth type	OAuth 2.0 (custom/partner app); scopes: read_orders, read_customers, write_orders.
Setup steps	User enters their store domain (yourstore.myshopify.com) → redirected to Shopify admin to approve scopes.
Triggers	New order created; Order fulfilled; New customer.
Actions	Add order note; Update order tags; Create draft order.
Sync mode	Webhook — native Shopify webhooks, registered automatically on connect; instant.
Notes	Uninstalling the connected app in Shopify auto-revokes the connection and pauses affected workflows.

12. Razorpay

Auth type	API Key ID + Key Secret (pasted from the Razorpay dashboard).
Setup steps	User pastes Key ID/Secret; system generates a webhook URL that the user adds manually under Razorpay → Settings → Webhooks (MVP is manual; auto-registration via Razorpay's API is a Phase 2 upgrade).
Triggers	Payment captured; Payment failed.
Actions	Create payment link.
Sync mode	Webhook (manually registered at MVP); instant once configured.

Notes	Webhook payload signature is verified against the Key Secret to prevent spoofed events.
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Group 6 — Scheduling / Productivity

13. Google Calendar

Auth type	OAuth 2.0 via Google (scope: calendar.events).
Setup steps	Google consent → select a specific calendar from a dropdown.
Triggers	New event created.
Actions	Create event; Update event; Add attendee.
Sync mode	Polling at MVP (1–5 min); push via Calendar watch channel is a Phase 2 upgrade.
Notes	Time zone is inherited from the selected calendar's own settings.

14. Calendly

Auth type	Personal Access Token (pasted from Calendly account settings).
Setup steps	Paste token → system validates it with a test call.
Triggers	New meeting booked; Meeting canceled.
Actions	None at MVP (source-only connector).
Sync mode	Webhook — auto-registered via Calendly's API on connect; instant.
Notes	Supports multiple event types on one account; user filters by event type when configuring the trigger.

Group 7 — Support

15. Freshdesk

Auth type	API key (from the user's Freshdesk profile settings) + helpdesk domain.
Setup steps	Enter Freshdesk domain and API key; system validates with a test call.
Triggers	New ticket created.
Actions	Create ticket; Add note to ticket; Update ticket status.
Sync mode	Webhook via a Freshdesk automation rule the user points at our URL (guided setup step); polling fallback every 5 min if skipped.
Notes	Chosen over Zendesk for MVP given lower cost and stronger SMB/India adoption; Zendesk is a Phase 2 add.

Group 8 — Generic Connectors (no third-party account required)

16. Webhook — Catch Hook

Auth type	None — we generate a unique, unguessable URL per workflow.
Setup steps	System generates the URL; user pastes it into any external tool/API/script that can POST JSON.
Triggers	Webhook received (any JSON payload; field list is auto-detected from the first test call).
Actions	N/A — trigger only.
Sync mode	Instant.
Notes	The universal fallback for any app not yet natively supported.

17. HTTP Request — Send

Auth type	Configurable per step: none, API key header, Bearer token, or Basic auth — no pre-connection needed.
Setup steps	User configures method (GET/POST/PUT/PATCH/DELETE), URL, headers, and body directly on the step.
Triggers	N/A — action only.
Actions	Call any external API endpoint; response is parsed as JSON for downstream field mapping.
Sync mode	Instant (synchronous call within the run).
Notes	Developer escape hatch used alongside the Code step (Phase 3) for anything not natively supported.

18. Scheduler

Auth type	None — internal to the platform.
Setup steps	User picks an interval (every X minutes/hours) or specific days/times and a time zone.
Triggers	Fires on the configured schedule.
Actions	N/A — trigger only.
Sync mode	Instant at the scheduled time (managed by the Trigger service, Section 11).
Notes	Used for digesting/report-style workflows that don't depend on an external event.

9. Key User Flows

9.1 First-run / Onboarding flow

7. Sign up → verify email.
8. Onboarding question: “What do you want to automate first?” (options mapped to templates, e.g. “Form → WhatsApp”, “WhatsApp → CRM”, “Start from scratch”).
9. If template chosen: pre-filled workflow opens in editor with placeholders highlighted — user just connects accounts and maps 2–3 fields.
10. Guided tooltip walkthrough: Trigger → Add step → Test → Turn on.
11. First successful test run triggers a celebratory confirmation and prompts “Turn this on now?”

9.2 Build-a-workflow flow (return user)

12. Dashboard → “Create Workflow”.
13. Choose Trigger app/event → connect/select account → configure → fetch sample data.
14. Add Action(s) — repeat configure/map/test per step.
15. Optional: add Filter/Router/Delay steps.
16. Name workflow → Publish (toggle On).
17. Land on Run History, empty state: “Waiting for the next trigger event...”

9.3 Error / failure recovery flow

18. A run fails at step 3 of 5.
19. Owner gets an email/Slack alert: workflow name, failed step, error summary, link to the run.
20. User opens run → sees failing step’s input/output → fixes the mapping or reconnects the app.
21. User clicks “Retry from this step” — steps 1–2 are not re-run; step 3 onward re-executes.
22. If retries repeatedly fail, workflow is auto-paused after a configurable threshold and owner is notified.

9.4 Upgrade / hitting task limit flow

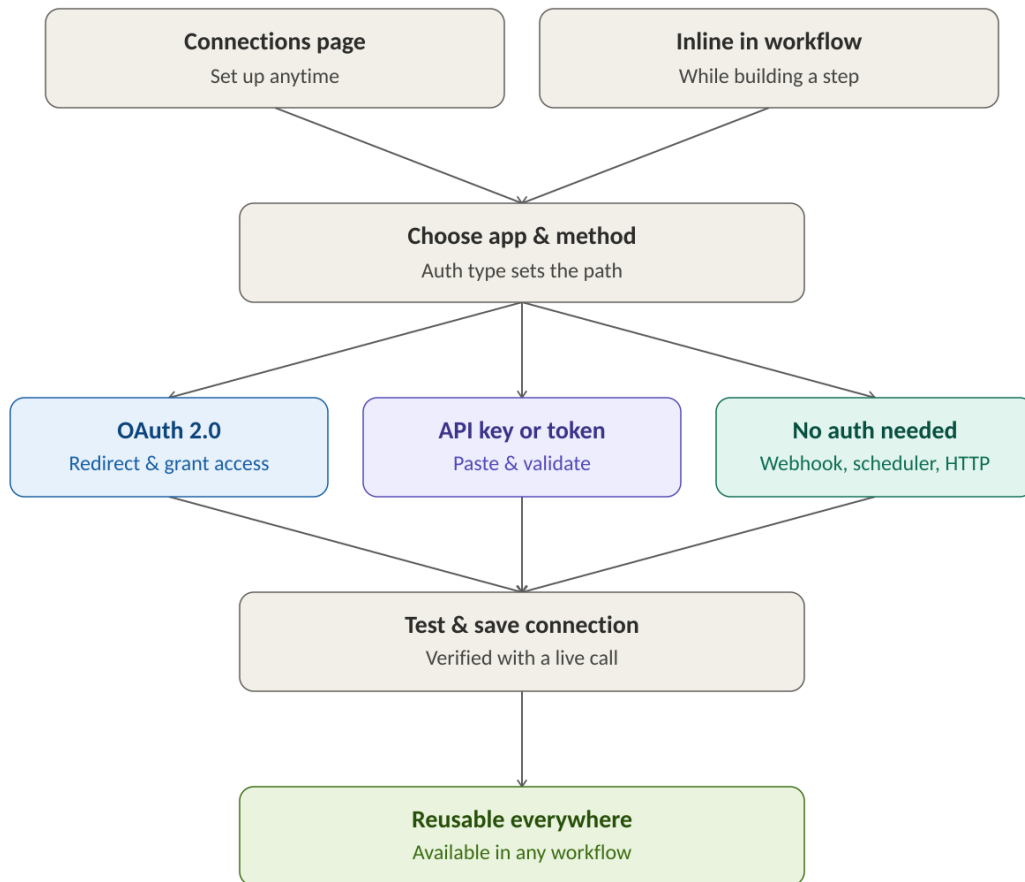
23. In-app banner at 80% of monthly task usage; email at 100%.
24. At 100%, new task-consuming runs queue (not silently dropped) and clearly show “Paused — upgrade to resume” in Run History.
25. One-click upgrade from the usage banner → plan picker → payment → queued runs resume automatically.

9.5 Team collaboration flow (Phase 2)

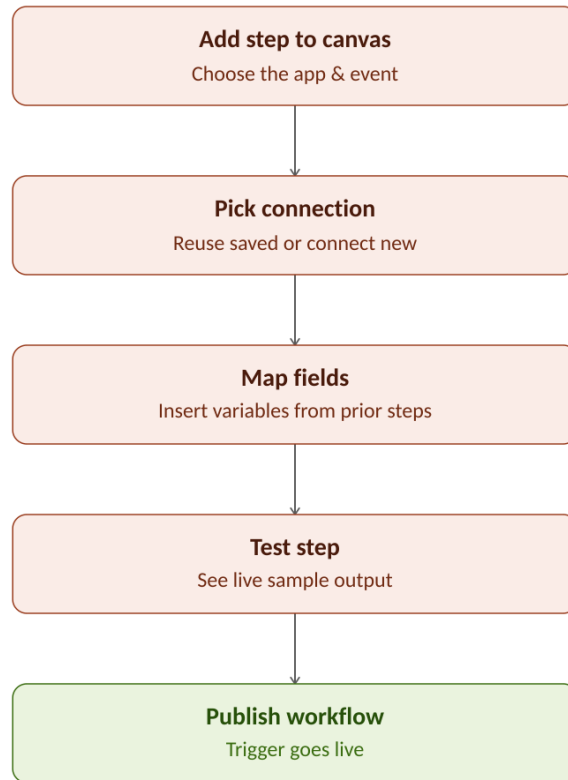
26. Owner invites a teammate by email with a role (Editor/Viewer).
27. Invitee accepts → sees shared folder(s) per their permission.
28. Editors can create/edit workflows in shared folders; Viewers can see run history but not edit steps or connections.

9.6 App connection flow (diagram)

How and where a user connects any of the 18 MVP apps (Section 8.8): two entry points — the Connections page or inline while building a step — both lead to choosing the app, whose auth type (OAuth, API key/token, or none) determines the setup path. All paths converge on testing and saving a reusable connection.

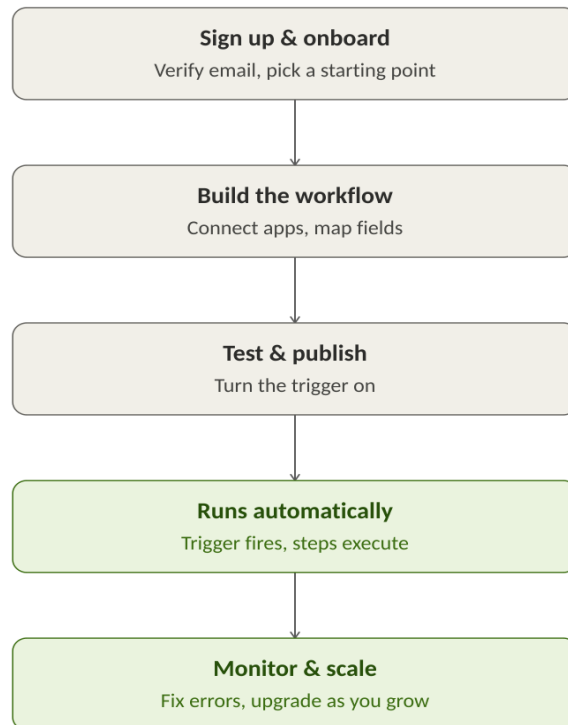


Once a connection is saved, this is how it gets used the next time the user adds a step to any workflow — either a brand-new connection or one already on file:

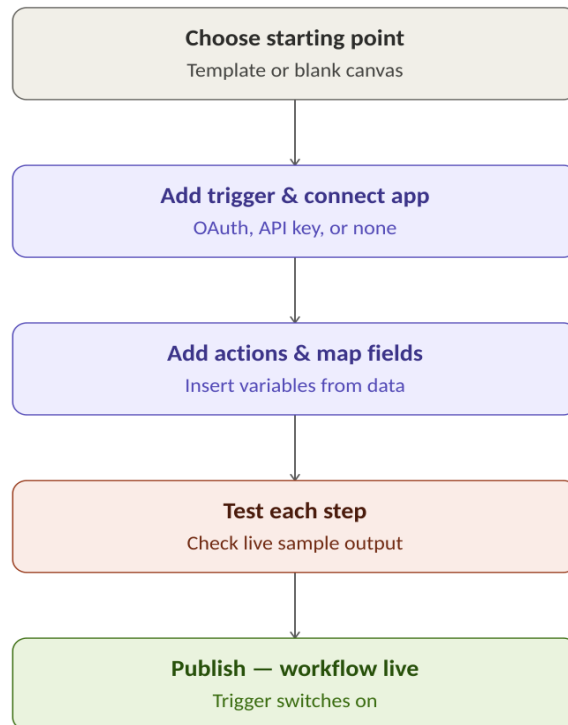


9.7 Complete app user flow (diagram)

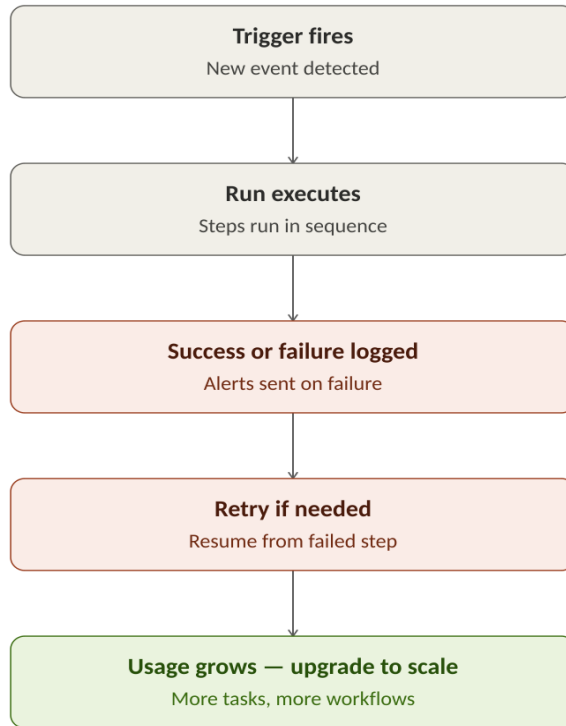
Zooming out from individual flows, this is the full lifecycle of a user in Automate Workflows, start to finish: sign up and onboard, build the workflow, test and publish it, watch it run automatically, then monitor and scale as usage grows.



The build phase (stages 1–3 above) expanded: choosing a starting point, adding a trigger and connecting an app, adding actions and mapping fields, testing each step, and publishing to go live.



And the runtime phase (stages 4–5 above) expanded: what happens each time the trigger fires, how successes and failures are logged and retried (Section 9.3), and how growing usage leads into the upgrade flow (Section 9.4).



10. Non-Functional Requirements

10.1 Reliability & performance

- Target 99.9% platform uptime from Phase 3 onward (99.5% acceptable for MVP/Phase 2).
- Webhook-triggered workflows start execution within 5 seconds of receipt (P95).
- Polling triggers check on an interval appropriate to plan tier (e.g., 5 min on Free, 1 min on Pro).
- At-least-once execution guarantee per run; steps are idempotent-safe where the destination app supports it.

10.2 Security & compliance

- All credentials encrypted at rest (AES-256) and in transit (TLS 1.2+).
- OAuth tokens refreshed automatically; expired/revoked tokens surfaced to the user, never silently retried indefinitely.
- Role-based access control from Phase 2; SSO/SAML and audit log from Phase 3.
- SOC 2 Type II and GDPR-readiness targeted ahead of enterprise (Phase 3/4) sales.

10.3 Scalability

- Execution engine horizontally scalable (queue + worker model) independent of the editor/web app.
- Per-tenant rate limiting to prevent one workflow from starving others.

10.4 Observability

- Full per-step execution logs retained per plan tier (e.g., 7 days Free, 30 days Starter/Pro, 1 year Business).
- Internal system dashboards for queue depth, error rate by app/connector, and task-consumption anomalies.

10.5 Accessibility & localization

- Editor meets WCAG 2.1 AA for keyboard navigation and screen-reader labeling on core flows.
- UI copy externalized for translation; English at launch, additional locales prioritized by user base (e.g., Hindi) post-MVP.

11. Technical Architecture Overview (High Level)

This is a product-level view for planning, not a final engineering design — engineering to own the detailed architecture doc.

- Editor/Web app — front-end SPA for building/managing workflows, connections, and viewing run history.
- Workflow definition service — stores workflow graphs (trigger + steps + config + mappings) with versioning.
- Trigger service — manages polling schedules and webhook ingestion endpoints per connection.
- Execution engine — queue-driven worker system that runs a workflow step-by-step, persists per-step input/output, handles retries.
- Connector layer — per-app adapters (auth, trigger polling/webhook, available actions, field schemas); designed so new connectors can be added without touching the core engine.
- Credential vault — encrypted storage for OAuth tokens/API keys, shared by trigger service and execution engine.
- Billing/usage service — meters tasks per successful external action, enforces plan limits.
- Notification service — email/Slack/in-app alerts for failures, usage thresholds, invites.
- AI service (Phase 4) — wraps an LLM to generate workflow drafts and power the in-workflow AI Action step.

12. Success Metrics

Metric	Why it matters	MVP target (illustrative)
Activation rate (created → published a live workflow)	Measures whether the builder is usable end-to-end	≥ 50% of signups within 7 days
Time to first live workflow	Onboarding effectiveness	< 15 minutes median
Workflow run success rate	Core reliability signal	≥ 98%
Weekly active workflows	Real usage vs. one-off setup	Track trend, no fixed target at MVP
Task-limit driven upgrades	Pricing/monetization health	Track conversion rate from 80%-usage banner
Support tickets per 100 active workflows	UX and reliability quality	Trend down release over release

13. Risks, Assumptions & Open Questions

13.1 Risks

- Building and maintaining connectors for many third-party apps is resource-intensive — mitigate by starting with a focused app list plus a generic Webhook/HTTP connector for anything not yet supported.
- Reliability expectations are high for automation tools — a few high-profile failed runs can erode trust; invest early in retry/alerting UX (Section 9.3).
- Pricing model (unbilled internal steps) is generous like Pabbly's — needs margin modeling before final pricing lock.

13.2 Assumptions

- Initial user base is existing Automate Chats / Automate Forms customers, so launch app catalog should prioritize those plus their most-requested adjacent tools (CRM, sheets, email, Slack).
- Desktop-first usage; mobile is monitoring/approvals only at MVP.

13.3 Open questions (need decisions before build)

- Final MVP app list — who owns prioritization: growth data, sales requests, or both?
- Exact plan pricing/limits (Section 8.7 is a draft placeholder).
- Build vs. buy for the connector framework (e.g., evaluate embedded iPaaS SDKs vs. fully custom).
- Brand/URL for this product within the Automate suite (e.g., automateworkflows.ai) — to align with [[automate-chats-landing-page]] and [[automate-forms-landing-page]] naming.

14. Illustrative Roadmap Timeline

Phase	Focus	Rough duration
Phase 1 — MVP	Core editor, ~15–20 apps, run history, basic billing	Months 1–4
Phase 2 — Growth	Router/Delay/Formatter/Loop, teams, 100+ apps	Months 4–8
Phase 3 — Scale	Code step, SSO/RBAC, audit log, public API, SLA reliability	Months 8–13
Phase 4 — AI & Enterprise	AI workflow generation, AI action step, data store, enterprise controls, embedding into Automate Chats/Forms	Months 13–18

Timelines are illustrative for planning discussion, not committed dates.